

Course Name: Multivariable Calculus

Credit Hours: 3 (3-0)

Contact Hours: 3-0

Pre-requisites: Calculus and Analytical Geometry

Course Introduction: To understand the basic concepts and know the basic techniques of differential and integral calculus of functions of several variables; To apply the theory to calculate the gradients, directional derivatives, arc length of curves, area of surfaces, and volume of solids; To solve problems involving maxima and minima, line integral and surface integral, and vector calculus;

Course Outline:

1. Functions of Several Variables and Partial Differentiation.
2. Multiple Integrals, Line and Surface Integrals.
3. Green's and Stoke's Theorem.
4. Fourier Series: periodic functions,
5. Functions of any period $P-2L$,
6. Even & odd functions,
7. Half Range expansions,
8. Fourier Transform;
9. Laplace Transform,
10. Z-Transform.

Teaching Methodology:

1. Lecturing,
2. Written Assignments
3. Class Assessment
4. Sessional Exam
5. Home Assignments
6. Quizzes
7. Mid term Exam
8. Final Exam

Textbook:

1. Multivariable Calculus, 6th edition James, Stewart 2007 Cengage Learning publishers.

Reference Material:

1. Calculus and Analytical Geometry, 6th edition. Swokowski, Olinick and Pence. 1994. Thomson Learning EMEA, Ltd.
2. Multivariable Calculus, 5th edition Howard, A. Albert, H. 1995, John Wiley.